

Lavoisier: Model-Based Systems Engineering in a Type System

An evidence-first account of modelling real-world systems in stable Rust, with conservation enforced by the compiler

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Abstract

Lavoisier is an experiment in using a statically-typed programming language — stable Rust, with no external dependencies beyond a test harness — as a medium for model-based systems engineering. Resources, requirements and processing states are types; processes are functions that take everything they need by value and return everything they produce; and a model that double-books a technician, loses mass between steps, overdraws a budget or uses a part no earlier step produced does not compile. The project was run evidence-first: twenty-two written requirements of the modelling approach itself (R1–R22), fifteen isolated experiments with pass/fail criteria (EXP-01..EXP-15), a consolidated findings log of sixty-four entries (F-001..F-064), a reusable library (`model-core`), nine downstream model crates — a validation pilot and a completed ladder of five case studies, from making a pot of tea to two-site fulfilment with procurement — an eight-step verification gate, and a generation-and-analysis toolchain that turns the models into diagrams, work instructions and lint reports, and turns machine-checked specifications into model scaffolds. This paper gives the full account: the method, the rules and the evidence behind them, what the compiler genuinely guarantees, what needs a layered regime of lints, tripwires and tests, the measured costs, and the limitations stated plainly.

Document status

Version	0.3 (the toolchain complete: generated documents, model analysis, the specification pipeline)
Date	2026-10-08
Covers	R1–R22; EXP-01..EXP-15; F-001..F-064; <code>model-core</code> ; the pilot workshop model; case studies CS-1..CS-5 (§6); the generated diagrams (<code>docs/diagrams/</code>), process documents (<code>docs/processes/</code>), analysis reports (<code>docs/analysis/</code>) and the specification gate (<code>tools/spec-gen</code>)
Status	Living document
v0.2 added	§6 (the case-study ladder, with CS-4’s change-impact evidence); F-055..F-057 and the 2026-10-07 extension notes folded into the findings synthesis; counts updated throughout
v0.3 added	§7 (PLAN steps 9–11: generated process documents, the model linter, the DSL experiment trio and the R22 specification pipeline); F-058..F-064 folded into §8; the gate grown to eight steps; counts updated throughout
v0.4 will add	Case studies born as machine-checked specs and <code>specgen</code> scaffolds, and whatever the open questions of §11.3 resolve next

Versioning convention. This is a living document, extended rather than rewritten. Each case study and each major deliverable (generated diagrams, process document generation,

the DSL, model analysis) adds sections and appends findings; existing sections are corrected but not restructured. Minor versions (0.2, 0.3, ...) mark those extensions; the document stays at 0.x until the open questions of §11.3 are resolved or formally parked. Every factual claim is verifiable against the repository: rule numbers (R), finding numbers (F) and experiment numbers (EXP) are cited inline and indexed in the appendices.

1 Motivation and hypothesis

Engineering plans and specifications are mostly written in prose, and prose hides contradictions. A technician booked on two jobs at once, a step that consumes a part no earlier step produced, material that silently disappears between operations, waste with no disposal route, a budget quietly overspent — none of these is visible on the page, because nothing on the page is obliged to balance. The errors surface later — in procurement, on the shop floor, at integration — where they are expensive.

The project’s founding observation is that this class of error is *bookkeeping* error, and bookkeeping can be audited mechanically. The name honours Antoine Lavoisier — “*Rien ne se perd, rien ne se crée, tout se transforme*”: nothing is lost, nothing is created, everything is transformed. If an engineering plan is written so that every resource — material, energy, time, money — must balance across every step, then an auditor that checks the books on every change will catch the contradictions at design time.

The specific hypothesis is that a modern type system can be that auditor. Rust was chosen because its ownership rules give the central property away: values move. A value passed to a function is gone from the caller; it cannot be used twice, and (with `Clone` and `Copy` forbidden on resources) it cannot be duplicated. If every *thing* in a plan is a type, and every *work step* is a function whose inputs and outputs must balance, then an impossible plan is a type error — and the compiler, software’s strictest proof-reader, rejects it with a message saying what is wrong and where. The project deliberately does not build software in the ordinary sense; it borrows software’s auditor.

Two refinements make this an experiment rather than a manifesto. First, the hypothesis was *expected to fail somewhere*: the stated aim (instructions.md, Summary) is “to find out how far a type system can be pushed for model-based systems engineering, and to record the limitations and complications that turn up along the way”. Where there was a choice, the project preferred the approach that puts more checking into the compiler, *even if the types get complicated*, precisely to find where that breaks down. Second, the claims are confined to consistency, not truth: the method proves a plan internally coherent — it does not prove the plan matches reality (a kettle might not really need 550,000 J), and it does not simulate performance. Validation against reality is an open question (§11.3), kept deliberately distinct from the verification this paper is about.

The constraint set matters to the result: **stable Rust only** (no nightly features such as `generic_const_exprs`), no dependencies except the `trybuild` test harness as a dev-dependency, and the project’s own units and type-level-number machinery rather than crates like `uom` or `typenum` (R8, R12). Every measurement in this paper was taken on `rustc 1.98.1` (stable, September 2026) on Apple Silicon macOS.

2 The method: an evidence loop

The project’s second experiment — alongside the technical one — was methodological: every design rule had to be backed by evidence before it was adopted, and every loss had to be

recorded rather than papered over. The loop ran as follows.

1. **Requirements agreed in writing.** The requirements of *the modelling approach itself* live in `instructions.md` as numbered rules — R1–R19 for the modelling method, R20–R22 for the toolchain that later grew around it (§7) — distinct from the REQ–NNN requirements of any modelled system. Changes to them are agreed between author and implementer before being committed; nothing is reinterpreted silently.
2. **Isolated experiments.** Each risky claim the rules make became one self-contained experiment crate (EXP-01..EXP-15), with build steps and pass/fail criteria fixed in advance; the first nine ran in parallel with no shared code, and the final three (the DSL candidates, §7.3) ran as one batch against the finished toolchain. Each produced a `RESULTS.md` with a per-criterion verdict (**pass** / **pass with complications** / **fail**), verbatim compiler output judged for how a modeller would read it, candidate findings, and an adopt/adapt/reject recommendation.
3. **A findings log.** A single consolidation pass turned the candidate findings into `FINDINGS.md` (R14): stable IDs F-001 onwards, each recording what could not be expressed (or only with complications), what it cost, and the workaround adopted, with evidence citations into the experiment crates. Later findings were added by the library and model builds themselves; a finding that is later resolved gains a dated `RESOLVED` note (the F-036 pattern) rather than being rewritten.
4. **Rules updated.** The experiment results were written back into R1–R19 as amendments — sometimes strengthening a rule (Peano numbers adopted for capacities, EXP-01), sometimes weakening or re-scoping one (the `Supplier` trait declared discrete-only, F-028), and three times creating a new rule outright (R17–R19 were open questions until EXP-10..12 answered them).
5. **Implementation.** Only then was the real library built: `model-core`, plus a pilot model (`pilot-workshop`) that exercises every feature, gated end to end by a CI script (§4.4).
6. **Case studies.** New models are written from natural-language specifications via `SPEC_TEMPLATE.md`, each chosen to stress something no earlier model did; CS-1 (making a pot of tea) completed the first full spec-to-model round trip (§4.5), and the full five-step ladder is now complete (§6).

Evidence-first mattered for three reasons. It kept the rules honest — every convention in §3 cites the finding that forced it, and several intuitive designs died in experiments before reaching the library (binary type-level numbers, F-012; the type-selected outcome token, F-042; the aggregate “workshop” parameter, F-024). It made the *negative* results durable: the findings log preserves what failed and why, with verbatim compiler output, so a future revisit does not re-run dead ends. And it surfaced genuinely unexpected results that casual development would likely have misattributed — most dramatically a deterministic compiler crash (F-034, §8.4) and the discovery that the compiler’s own suggested fixes systematically recommend conservation violations (F-007, §5.2).

3 The approach: R1–R19 by theme

The nineteen modelling rules are stated normatively in `instructions.md`; this section synthesizes them thematically. (The three toolchain rules that followed them, R20–R22, arrived with the generation and analysis deliverables and are the subject of §7.)

3.1 Conservation and ownership

The core rule, R1, is Lavoisier’s principle operationalized: a process function transforms inputs into outputs with no side effects — it may neither create resources from nothing nor consume an input without turning it into some output (product, by-product or waste). Everything a process needs is moved in by value; everything it produces is returned. R2 sharpens this into strict exclusivity: resources are never borrowed or shared (no `&T`, `&mut T`, `Rc` or `Arc` in process signatures); a reusable resource — a person, a tool, a location — is moved in and *returned*. The compiler consequently enforces one resource in one process at a time (F-025), and a double-booked person fails with an error pointing at both competing uses.

“No creation from nothing” is made structural rather than conventional: resources have private constructors (sealed types with at least one private field), so only the boundary code that represents real-world arrival may mint them (R1, R12). Changing an amount is itself a process (R3), with the balance checked by a compile-time assertion (`const { assert!(IN == A + B) }`) — splitting 2000 g into 1500 g + 600 g is a compile error.

The known gap is stated in the rules themselves: Rust’s types are affine, not linear, so a value can be *dropped* silently. “Nothing is lost” is therefore only partly a compile-time property; the remainder is covered by a layered regime (R1, quantified in §8.3): `#[must_use]` on every resource with the deny-lints that back it, a panicking “tripwire” destructor on every consumable resource that converts silent test-time leaks into named failures, a clippy configuration banning the escape hatches, and per-branch test coverage. The rules are explicit that the set is only sound as a whole, and that the residual gaps are accepted and documented rather than denied.

3.2 Quantities, units and capacity

Every quantity and unit is a type that also carries its value (R7): integers in base units — grams, millimetres, milliseconds, square and cubic millimetres, millikelvin, milliamperes, joules, and minor currency units — with no floating point. Unit mixing is a type error: grams cannot meet millimetres, and (by R19) pence cannot meet euro cents. The project writes its own units library (R8) rather than depending on existing crates (§10).

Discrete items are separate objects, never counts (R13): a bolt is a value, and a box of bolts *contains* its bolt values. Capacity is type-level (R12): the box’s remaining contents are a type-level list whose length is the count — a box of 100 bolts is literally a type one hundred levels deep, on purpose — so an exhausted supplier simply does not implement `Supplier`, and drawing a fifth bolt from a box typed as holding four is a compile error. The type-level numbers behind this are the project’s own Peano naturals, adopted over a binary encoding by experiment (EXP-01, F-012), with measured guidance: capacities to ~100 are free, ~500 is the comfort ceiling, and every participating crate sets `#![recursion_limit = "2048"]` (F-010).

Continuous material takes the opposite representation (R15): a sealed container with a const-generic magnitude (`GasBottle<5000>`, `Battery<E>`, `Account<BALANCE>`), drawn down by an explicit draw process in which the caller states the remainder and a const assert checks the arithmetic. Overdrawing is then a conservation violation with *no compiling remainder* — finite capacity falls out of conservation for free. Rates and wall-clock time stay outside the model (R9, R15): what is conserved is the amount — joules, grams, person-milliseconds — not watts or schedules. Time as a costed input is a budget on the resource that spends it (`Person<300_000>`), drawn down like the gas bottle.

3.3 States and characteristics

A thing’s characteristics are encoded in types (R6), two ways with fixed roles settled by experiment (EXP-05): catalogues of orthogonal values are parameterized types (`Bolt<Size, Material, Length>`, each slot policed by a “kind” trait so transposed arguments fail at the construction site, F-018), while requirements and capabilities are marker traits, bridged from the catalogue by one-line blanket impls (F-019). Each processing state is its own type (R9): a `Plate`, a `DrilledPlate` and a `ScrapPlate` are three types, converted only by processes. This is what makes “product never produced” a compile error — “expected `DrilledPlate`, found `Plate`” reads directly as *these plates have not been drilled yet* (F-023) — and the same rule extends to failure states (a broken drill bit is not a drill bit) and safety states (a fitted guard is not a loose guard, R18).

R9 also fixes what the model is *about*: connections, not sequences. Process signatures declare which outputs feed which inputs; any order satisfying the data dependencies type-checks, and independent branches may run concurrently because strict conservation proves they share no resource. The model deliberately contains no clock.

3.4 Requirements and traceability

Requirements of the modelled system are first-class artifacts (R10): each is a named trait whose documentation begins with its id (`REQ-NNN`), composed of characteristic supertraits with a blanket impl, and used as a bound on every process that demands it. Because the blanket impl makes satisfaction invisible to text search (F-020), each satisfying type carries a `/// Satisfies: REQ-NNN` doc tag *backed by a compile-checked assertion* — the tag is for grep, the assertion is for truth, a stale tag a compile error naming the missing characteristic. Tests are tagged `/// Verifies: REQ-NNN` (R5 requires at least one test per process, and per `Result` arm for fallible processes). A dependency-free shell script, `trace.sh`, turns the source into a traceability report — requirement → defined at / satisfied by / verified by — and fails CI on any warning, including a requirement with no verifying test. The discipline that keeps grep reliable (exact tag case, tags only in `///` doc comments, bounds on the `fn-name` line, no tags inside macro invocations) is itself linted by the same script (F-021, F-037).

3.5 The boundary

Everything enters the model through explicit suppliers and leaves through explicit consumers (R12), which are themselves resources, passed by value, with finite, type-level capacity — and which are *strict*: a supplier takes nothing but itself and yields one item plus its next state; anything that needs payment or paperwork is a process chain, not a supplier. Repeated use goes through a recursive helper trait (`SupplyN<N>`), so “take four bolts” costs one where-clause (F-014). Continuous sources are draw-style boundary processes, not `Supplier` impls — the trait is provably discrete-only on stable Rust (F-028).

Practically unbounded environments — the atmosphere, mains water, the grid — are `Next = Self` boundary objects, legal *only* at the boundary (inside the model they would mint or swallow resources, exactly what R1 forbids) and always placeholders. The placeholder convention (R12) is the model’s named ignorance: a greppable `/// Placeholder:` tag on every assumed supplier, sink, outcome source and exchange rate, each a designed refinement point. Consumed time and other events worth recording go to **History** (R16), a sealed boundary consumer that defuses what it consumes but keeps a value-level record — “the past” as an explicit sink, doubling as the execution record. Concurrent branches carry separate histories merged at joins into an honest partial order: no interleaving is invented.

3.6 Variability, people and money

The last three rules were open questions answered by the final experiment batch. **Fallible processes** (R17, from EXP-10): a process that can fail returns `Result<OkBundle, FailBundle>` where *both* arms conserve the same inputs — a scrapped part is a product, not a disappearance — with one `const assert` per branch, both checked at every call site. Variability enters only at the boundary as a sealed, runtime-valued outcome token that flows cannot read; the type-selected alternative was built and rejected because it collapses fallibility back into two infallible processes (F-042). Outcome bundles deliberately implement no `Debug`, which makes `.unwrap().expect()` — the panicking shortcut past the failure arm — a compile error (F-047). Rework is bounded by provisioning: a retry consumes provisioned reserves, so unbounded retry is inexpressible, which is the honest statement of real rework (F-050).

Qualifications and safety (R18, from EXP-11): certifications are marker traits on a sealed `Qualified` wrapper around the common `Person` — never on `Person` itself, where a single `impl` would certify everyone in the model at once (F-043) — and safety states follow one-type-per-state, so an unqualified operator and an unfitted guard are distinct, REQ-phrased compile errors. **Money** (R19, from EXP-12): each currency is its own dimension in integer minor units; cash and accounts reuse the R15 container machinery unchanged, so overspending is the standard overdraw error; vendors implement `Consumer` only at their exact price, which moves wrong payments to type-check time with the correct price named in the error (F-051); and currency exchange — value-equivalence at a stated integer rate, not conservation — is confined to the boundary, `const-asserted` exactly, so silent rounding is unrepresentable (F-052).

3.7 The verification gate

R4 and R5 close the loop: compile errors must expose model mismatches, with pinned compile-fail regressions, and unit tests must verify every transformation. The crucial caveat — the project’s single most consequential finding (F-001, §5.1) — is that conservation asserts fire at monomorphization, during code generation, so `cargo check` and the editor diagnostics built on it cannot see them. The gate (§4.4) is therefore built on full builds, never on `cargo check` alone.

4 Implementation

4.1 model-core

The library (`model/model-core`) is deliberately small — eight public modules and a feature-gated fixtures module:

- `nat` — Peano type-level naturals with generated aliases `N0..N1000` (the alias table is written by a shipped generator script, because stable `macro_rules!` cannot synthesize identifiers, F-013);
- `list` — type-level lists of real values, whose length *is* the count;
- `quantity` — the `Unit` kind trait, the eight base units, sealed `Qty<V, U>`, and conserving `split/combine`;
- `resource` — the sealing/tripwire kernel (§4.2);
- `boundary` — `Supplier` (discrete-only), `Consumer`, recursive `SupplyN/ConsumeList`, and the generic access processes, all carrying modeller-phrased `#[diagnostic::on_unimplemented]` messages;

- **common** — Person with an R15 time budget, the Qualified wrapper, Organisation, Location, Labour and its production-legal TimeLedger sink (F-035);
- **history** — the R16 execution history;
- **requirement** — the R10 pattern as macros compatible with `trace.sh`.

`model-core` is infrastructure: it defines no REQ-NNN requirements itself. Its crate documentation carries the conservation obligations every modelling crate must adopt (recursion limit, `forbid(unsafe_code)`, `deny-lints`, tripwires) and a condensed error-reading guide — because the sealing pattern is “one attribute away from broken” with no compiler warning (F-005), the rules are stated where every modeller will read them.

4.2 The kernel macros

The hard-won part of the library is boilerplate generation. Sealing a resource correctly takes about ten lines of tripwire machinery plus the sealing checklist per type (F-008); the `resource` module’s macros (`consumable_resource!`, `container_resource!`, `reusable_resource!`, `draw_process!`, `outcome_token!` and friends) generate the sealed struct, the `#[must_use]` attribute, the tripwire `Drop`, the sanctioned defuse site, boundary constructors and test-support fixtures together, so a downstream resource is one declaration. The macros initially accepted only non-generic types (F-036); extending them to generic parameters and payload fields required a token-munching parser with documented grammar limits, and surfaced its own finding (F-040: tripwired wrappers must hold *untripwired* contents, or defusing the wrapper would erase the contents’ leak protection). EXP-12’s verdict is the pay-off measured: modelling money needed **zero** new library machinery — `container_resource!` and `draw_process!` fit cash and accounts verbatim (F-051).

4.3 The models

pilot-workshop is the validation model, built alongside the library and exercising all of it: a steel sheet is cut into plates, drilled (a certified operator behind a fitted guard, drawing time from a budget; drilling can fail, scrapping the plate or snapping the bit), and fastened into an assembly with four catalogue bolts from a BoltBox supplier; swarf goes to a contents-keeping bin with a sealed disposal path (F-039); labour is recorded into per-branch histories merged at the join; money pays for parts through an exact-price vendor. Five requirements (REQ-001..REQ-005) are defined, satisfied and verified through `trace.sh`.

cs1-pot-of-tea is the first *case-study* model — built not alongside the library but from an agreed natural-language specification (§4.5). Seven further case-study crates followed it down the same workflow — `cs2-puncture-repair`, `cs3-cafe-orders`, the two-crate `cs4-stores/cs4-line` pair and the three-crate `cs5-supply/cs5-logistics/cs5-works` split — bringing the workspace to ten crates and twenty-seven modelled-system requirements (REQ-001..REQ-027, allocated workspace-uniquely per F-053). The ladder itself is §6.

4.4 The CI gate

`model/ci.sh` is the whole verification story in one command, eight steps, each existing because a finding proved it necessary:

1. **Full cargo build** — conservation E0080s fire at monomorphization; a gate resting on `cargo check` would pass violating models (F-001).
2. **cargo test** — unit and integration tests, `trybuild` compile-fail cases, tripwire `should_panic` demonstrations, and the `rustdoc compile_fail` doc-tests that are the

- only automated regressions for conservation violations (trybuild cannot see them, F-003).
3. **Clippy under -D warnings** plus the explicit restriction lints — the conservation lint set is only sound as a whole (F-007).
 4. **A plain no-features build** of every member’s production targets — during `cargo test`, feature unification compiles production sources against the test-support-featured library, so only this build proves production code cannot reach fixture constructors (F-004).
 5. `trace.sh` — the traceability report; any warning (including an unverified requirement) exits nonzero.
 6. **A feature-placement audit** — a `grep` proving test-support never appears under any [dependencies]-like section.
 7. **The model-analysis gate** (R21, §7.2) — `tools/lint.sh --check` regenerates the linter’s reports and diffs them against the committed `docs/analysis/` (stale reports fail), then fails on any gate-fatal ERROR finding.
 8. **The specification gate** (R22, §7.4) — `tools/spec.sh --check` validates every `case-studies/*/SPEC.md` against the template’s machine conventions; this error class fires before any code exists (F-062).

The model workspace currently carries 164 test functions plus 45 pinned trybuild compile-fail cases and the rustdoc `compile_fail` doc-test regressions — over two hundred automated checks run end to end on every change.

4.5 The spec-to-model workflow, and CS-1 as the worked account

`SPEC_TEMPLATE.md` defines how a model begins: a natural-language specification whose sections map mechanically onto model elements — §2 requirements become REQ-NNN traits verbatim, §3 resources become sealed types with their kinds (discrete/continuous/reusable/product), states and quantities, §4 the boundary table (every waste in §5 must have a destination row), §5 processes with their balance lines, §6 flows as connections with the concurrency stated, §7 assumptions destined for Placeholder: tags. The template’s governing rule: *anything the specification leaves out does not silently default — it comes back as a numbered question*.

CS-1, making a pot of tea, ran this loop in full. The spec (v0.2 agreed, v0.3 as implemented) models one person, an electric kettle, a teapot and a box of 40 teabags inside a kitchen boundary: five processes (fill, load, boil, pour-and-brew, empty the bin), with water (1500 g), energy (550,000 J drawn; 500,000 J embodied in the boiling water, 50,000 J kettle losses) and time (75,000 ms across four draws from a 300,000 ms budget) conserved per process — the brew’s mass balance is $1500 + 9 = 1473 + 36$ — and one genuine ordering freedom: the kettle boils with *no person*, so loading the pot before or during the boil are both valid orders, and both compile as separate integration flows. Four requirements (boiling water only; exactly 3 teabags; spent bags to the food-waste bin; waste heat accounted to the kitchen air) are enforced as REQ traits; the implementation produced one new positive pattern (F-054: a requirement-bounded process can consume a state-changing resource and keep REQ-phrased errors, via associated-const magnitudes and a permit-gated extraction).

The round trip was itself instrumented: implementation sent back eight numbered feedback items — among them, that the template should state where a time draw lives (adjacent process, F-048), should ask whether a process takes its consumers as parameters or the flow routes its outputs, and should distinguish *assert* from *structural* balance lines — all folded into the template at the review gate. Most significantly, the workflow found a real tooling flaw:

CS-1’s spec numbered its requirements REQ-001..REQ-004, but `trace.sh`’s id namespace is workspace-global and the pilot already owned REQ-001..REQ-005 — implementing the spec literally left the CI gate **green** while the traceability report silently merged CS-1’s tags under the pilot’s requirements (F-053), the dangerous green-but-wrong direction. CS-1 shipped as REQ-006..REQ-009 with the mapping documented, and the template now instructs spec authors to allocate workspace-unique ids. That the first real use of the workflow surfaced a silent hole in the verification tooling — and closed it — is itself a result: the method audits its own instruments.

5 The experiments

Each experiment tested one risky claim in isolation, with pass/fail criteria fixed in advance. The table gives one row each; the subsections discuss the decisive ones.

EXP	What it tested	Verdict (headline)	Key fact
01	Type-level naturals at useful sizes (R12)	pass w/ complications	Peano adopted; <code>recursion_limit</code> $\approx N+3$; 1.0 s build at $N=100$, 4.6 s at $N=1000$; binary flat ~ 1 s but ~ 25 impls vs 7 and no less-than
02	Suppliers/consumers as type-level lists (R12)	pass	<code>SupplyN</code> makes N draws one where-clause; capacity 100 builds in ~ 2.6 s; 130 breaks the default recursion limit
03	How close to “nothing silently lost”? (R1)	fail at compile time; pass as layered regime	8 leak paths; 3 have no compile-time detection; tripwire converts 6 of 8 into test failures
04	Compile-time conservation of quantities (R3, R7)	pass w/ complications	Violations are genuine E0080s — but post-monomorphization: <code>cargo check</code> , <code>rust-analyzer</code> and <code>trybuild</code> all pass them
05	Marker traits vs type parameters (R6)	pass	24-combination catalogue: 33 lines parameterized vs 121 hand-written markers; adding a value: 2 lines vs 6+
06	Mechanical traceability (R10)	pass	~ 150 -line dependency-free <code>trace.sh</code> ; zero false positives/negatives over 11 edge cases
07	Flows, exclusivity, reusable threading (R2, R9)	pass	Double-booking errors point at both uses; a process that swallows a tool compiles at its own definition site

EXP	What it tested	Verdict (headline)	Key fact
08	The privacy boundary across crates (R1)	pass w/ complications	No safe-Rust hole in sealed types; seven one-attribute-away holes catalogued; cargo test alone cannot prove the fixture boundary
09	Continuous resources, budgets, sinks (R15)	pass	The whole R15 combination composes; errors print decimal magnitudes; Supplier proved discrete-only
10	Fallible processes (candidate R17)	pass w/ complications	Both arms conserve, checked at every call; <code>.unwrap()</code> does not compile; cost: 8 const parameters on one process
11	Qualifications and safety (candidate R18)	pass w/ complications	REQ-phrased refusals (“this person may not drill...”); wrapper ~34 lines once vs ~40 duplicated lines per parallel person type
12	Money as a conserved dimension (candidate R19)	pass	Zero new library machinery; exact-price vendors make wrong payments <i>editor-visible</i> ; exchange is integer-exact
13	DSL route 1: a model! macro front-end (§7.3)	reject	Curated errors survive (tokens carry spans) — but the model becomes invisible to every extraction tool: the lint gate passes vacuously green
14	DSL route 2: an external notation compiled to Rust (§7.3)	adapt (parked)	Both error layers and the whole tool surface survive; ~4× denser than the Rust it replaces, over the continuous core only
15	DSL route 3: generation from the SPEC document (§7.3)	adapt (adopted as R22)	The spec mechanically determines ~half a model — the structure, none of the error machinery; 12 enumerable holes; spec validation is its own pre-code error class

5.1 EXP-04: where conservation checking actually lives

The project’s central promise — “mass in = mass out, checked by the compiler” — is true, with an asterisk that shapes everything else. On stable Rust the check is an inline `const { assert!(A + B == IN) }` in a generic function; a violation is a genuine, unavoidable E0080 — but it evaluates only when a fully concrete instantiation is reached *during code generation*. Verified directly: compiling the violating program with `--emit=metadata` (what `cargo check` does) succeeds; a full build fails. So editor diagnostics report violating models as fine; `try-build`, which runs `cargo check` internally, reports “Expected test case to fail to compile, but it succeeded”; and generic code never instantiated is never checked (F-001, F-003). The whole verification architecture — `ci.sh` gating on builds, conservation regressions as `rustdoc compile_fail doc-tests`, R5’s every-process-instantiated test discipline, and the documentation warning that *while you type, your editor is lying to you about balances* — follows from this one measurement.

5.2 EXP-03: the affine gap, quantified

EXP-03 catalogued eight ways a resource can silently die (end-of-scope drop, `let _ =`, shadowing, `mem::forget`, `mem::drop`, `struct-pattern .. drops`, early return, panic unwinding) and crossed them with every stable counter-mechanism. The matrix is the project’s honest boundary: three paths have no compile-time detection by any stable mechanism, and panic unwinding evades everything (a tripwire that stays armed during unwind double-panics and aborts the whole test binary — worse than the leak). The adopted regime — `must_use` + deny-lints at compile time, the tripwire `Drop` at test time, `clippy`’s banned-methods list, branch-coverage tests — converts six of the eight paths into failures. The experiment’s most striking single finding is F-007: each mechanism’s official fix-it suggestion is the *next leak path* (`unused_must_use` suggests `let _ =`; `let_underscore_drop` suggests `drop(...)`; for a swallowed parameter `rustc` suggests borrowing, which R2 forbids). A subset of the lints gives false safety, and modellers must be explicitly taught to ignore the compiler’s advice on this axis.

5.3 EXP-08: the boundary holds, under discipline

Sealed types showed no safe-Rust hole from downstream — but seven one-attribute slips silently reopen the boundary (`#[derive(Default)]`, `#[derive(Clone)]`, public fields, a pub unit struct, a pub `enum`, any public data-to-Self function, and `unsafe`), each demonstrated by a passing test from the user crate. The sealing rules exist because the compiler will not defend the pattern against its own author; each rule is mechanically greppable. The experiment also settled test-fixture mechanics: `#[cfg(test)]` helpers can never serve downstream tests, and the `test-support` feature alternative leaks into production sources during `cargo test` via feature unification — measured with a probe that `cargo test` accepted and plain `cargo build` rejected (F-004). Hence `ci.sh` steps 4 and 6.

5.4 EXP-10..12: the open questions close

The final batch ran against the finished `model-core` and converted the three open questions into R17–R19. Two results stand out. First, a negative with teeth: selecting a process’s outcome with *types* is fully expressible on stable Rust — and useless, because the outcome becomes part of the flow’s static text and nothing ever forces the unchosen arm to be handled; fallibility requires exactly one runtime-valued sealed token per process kind (F-042). Second, an unplanned positive: because outcome bundles hold sealed resources and derive no `Debug`, `Result::unwrap()` and `.expect()` — the universal panicking shortcuts — simply do

not compile (E0277), closing at type-check time a hole the design had expected to police by convention (F-047).

6 The case-study ladder

With the library finished, five case studies ran the spec-to-model workflow end to end — each from an agreed SPEC_TEMPLATE.md specification (with its review decisions and implementation round-trip feedback recorded in the spec itself), each its own downstream crate (or crates) in the workspace, each through the full gate (then six steps; eight today) with zero trace warnings, and each chosen to stress something no earlier model did. The ladder is complete: eight case-study crates beside the pilot, REQ-006..REQ-027 continuing the workspace-unique sequence, and findings F-055..F-057 plus seven dated extension notes raised along the way.

6.1 CS-1: making a pot of tea

The smallest complete system, and the first spec→model round trip (§4.5): continuous resources dominate (water in grams, energy in joules, time drawn from a person’s budget), with one genuine ordering freedom (the kettle boils with no person) compiled as two separate flows. It stressed the *workflow* as much as the method — and the workflow caught a silent hole in the verification tooling itself (F-053, workspace-global requirement ids) and produced the permit-gated extraction pattern that keeps requirement-phrased errors on consuming processes (F-054).

6.2 CS-2: bicycle puncture repair

Fallibility in anger: a patch can fail, and the rework is provisioned — two patches and two outcome tokens in the kit make a third attempt *inexpressible* (R17/F-050). The build extended F-050 with a sharper observation: provisioning also polices *when* contingency resources may be acquired — buying the spare tube before the second failure is known has no compiling flow, because on the success arm the unused spare has no consumer left. Qualification ran as workshop induction (R18), money as the counter purchase (R19), and the deliberately multi-type requirement (REQ-012: patched-and-checked *or* new tube) turned F-055’s predicted many-to-many ambiguity into a real silent diagram error, fixed in the generator the same day.

6.3 CS-3: café order fulfilment

Genuine concurrency at last: two staff, two branches sharing no resource, and R16’s per-branch Histories merged at the join into an honest partial order — asserted to have the Join shape, with no invented interleaving, on all three paths (first-try, re-steamed, refunded). The merge machinery passed its first real two-actor exercise with no model-core changes (F-041 extended). A trybuild case pins the contention counter-example (“use of moved value” = this server is already busy), and the build added a sixth F-055 gap: statement-order freedom can hide flow structure from tools, so the convention is now to match directly on a fallible call.

6.4 CS-4: small-batch production run

The scale-and-structure case study, and the one with the most evidential weight: 25 assemblies through a 25-cycle type-level batch recursion (a loop cannot express it — the types change every cycle), a 100-bolt box, a 75-item swarf bin, and the first **two-crate model** —

cs4-stores owns every material type and seal, cs4-line can create nothing, so REQ-019 (provenance) is enforced by the crate edge itself, the EXP-08 result finally carrying a real model.

Its centrepiece is the **change-impact exercise** (`case-studies/cs4-batch-run/CHANGE-IMPACT.md`), turning “change a type and the compiler enumerates every affected process” from claim into measurement. Two probes were applied in stores and reverted exactly. **Probe A** (a characteristic change: the issued bolt’s length, L15 → L18) produced a blast radius of 2 errors in 2 waves across the 2 crates — the stores-side compile-checked catalogue-spec assertion, then the line’s bolt-spec bound at the exact fasten call inside the batch cycle — every error an E0277 phrased by the marker’s `on_unimplemented`, with the satisfying `impl` named, and all of it visible to `cargo check` and editors. **Probe B** (a quantity change: the sheet mass, 5000 g → 4800 g) also radiated 2 errors in 2 waves, but through conservation asserts instead: first a stores-side top-level batch-arithmetic `const` item, then the one unbalanced in-process assert, located by the “while instantiating `fn shear_sheet:<4800, 2250, 2250, 500>`” note with real decimal magnitudes. The two waves split exactly on F-001’s boundary — and refined it: wave 1 is check-visible, because a *non-generic* top-level `const _: () = assert!(...)` item evaluates at check time, so hoisting whole-model arithmetic into concrete `const` items recovers editor visibility that in-process generic asserts cannot have; wave 2 is invisible to `cargo check`, demonstrated directly (exit 0 under a violating workspace).

Two structural results surfaced only at two-crate scale. First, **the enumeration arrives in per-crate waves**: `cargo build` stops at the first failing crate, so the radius is enumerated iteratively — acknowledge wave n to see wave $n+1$ — and nothing downstream can be silently missed, because the downstream crate cannot compile until its wave is addressed. Second, **generic processes collapse repetition out of the radius**: 25 unrolled cycles contribute one error per distinct source location, not twenty-five. Within what the types state, the enumeration was complete — every affected process was named — while prose mentions of “15 mm” and the *new* split for 4800 g remain the human’s to re-derive (F-022/F-030, unchanged at scale).

The scale measurements landed on the comfortable side: the full 25-cycle composite recursion compiles in ~ 1.0 s (F-011’s flat region holds at real composite depth); the whole cold gate grew 44.9 s → 45.5 s (+1.4%) when the two crates joined; measured recursion-limit minima are ≈ 151 (stores) and ≈ 143 (line), so the mandated 2048 leaves $\approx 13\times$ headroom — with the new wrinkle that integration-test crates need the attribute too (F-010 extended). One design casualty became a finding: a `const` budget cannot descend through type-level recursion (`BUDGET - 150_000` per cycle needs nightly’s `generic_const_exprs`), so the operator’s shift is a **type-level quantum clock** — time held as 150 quanta of 30 000 ms, decremented structurally per fixed-size draw, each draw minting conserved Effort in real milliseconds (F-056). The costs are quantized draws and an ~ 40 -line parallel labour family; the consolation is that overdraw errors become check-time trait failures — editor-visible, unlike R15’s E0080s.

6.5 CS-5: two-site fulfilment with procurement

The boundary-refinement capstone, split across **three team-shaped crates** (`cs5-supply`, `cs5-logistics`, `cs5-works`; the `works` depends on both). R19’s last unexercised clause finally ran: two currencies as separate dimensions, bridged only by the bureau’s integer-exact boundary exchange (2000 p → 2340 ec at 117/100; a non-exact amount would not compile, F-052). The R12 evolution path was walked for real: the vendor — an unbounded placeholder in every earlier model — is refined into a modelled organisation with a *finite* three-component stock (a fourth purchase is a compile error), deliberately beside the bureau and customer, which stay placeholders for contrast. Transport is one type per location-state

threaded through a contracted courier organisation (REQ-027).

Crossing crate boundaries with sealed money forced the two patterns of F-057: **the holder wraps** (the works' Account<P> holds supply's sealed Money<P> as payload, so the balance parameter and the held cash cannot drift) and **the currency owner mints** — a crate that owns a currency must ship parameterized boundary entry/exit functions, since downstream boundary objects cannot mint sealed cash. The build also found the limit of CS-1's permit pattern: permit-gated extractions are same-crate only, so cross-crate consumption works by *keeping* (the resource as untripped payload inside the product), with the sealed characteristic living beside the type in the owning crate and the requirement trait living with whoever states the requirement (F-054 extended).

7 From model to paperwork, and back

With the ladder complete, the programme's last three planned deliverables (PLAN steps 9–11) turned the models into the documents engineering actually runs on — and turned the documents back into models. All three follow one architectural decision: every tool is std-only, parses ordinary committed source (or the specification document itself), shares one extraction library, and is forbidden to guess — what it cannot verify it states as a warning. The step-10 experiments made that decision a rule (R22). Three new rules, R20–R22, entered `instructions.md` by the same agreed-before-committed route as the first nineteen.

7.1 Generated process documents (R20)

`tools/docgen.sh` (the `docgen` binary of `tools/diagram-gen`, beside the diagram generator) writes one human-readable document per process, boundary function and composite flow — 166 documents across the nine model crates, committed in `docs/processes/` and regenerated after any model change. Nothing in them is invented prose: the signature is the work-instruction skeleton, the `const` parameters and named `consts` supply the numbers in base units, `Placeholder:` tags are the open-items list, the requirement bounds with their `Verifies:` tags are the compliance matrix, and the R9 connection graph states where each process fits as *needs / feeds* — a dependency graph, never a fixed procedure. The structured doc-comment fields once anticipated (`/// Actor:`, `/// Duration:`) proved unnecessary: signatures, adjacent draws and History records already carry that data. Building the generator also closed two of F-055's tooling gaps in the shared scanner — supplier items are now recovered through four concrete shapes (gap 1), and workspace path-dependencies are loaded transitively, so multi-crate models come out whole with cross-crate edges (gap 7); `stderr` warnings fell 42 → 6.

7.2 The model linter (R21)

`tools/lint.sh` (the `modellint` binary, sharing the same extraction library — never a fork) unifies the completeness and style checks previously scattered across `trace.sh` warnings, sealing greps and prose-only conventions: 23 catalogued checks (11 ERROR, 7 WARN, 5 INFO), each citing the rule or finding it enforces, written as one committed report per crate in `docs/analysis/`. The severity ladder is itself a designed artifact: ERROR is gate-fatal; ERROR-KNOWN is a maintainer-acknowledged true positive, named in the tool with its justification and never silently retuned; WARN is a should-fix convention; INFO is the improvement surface (placeholder density, thin verification, wide processes). `ci.sh` step 7 regenerates the reports and diffs them against the committed set, so a stale report fails the gate.

Its first run over the ten-crate workspace is a result worth stating in prose. R1’s sealing rules held across every crate with exactly one systematic exception: the three contents-keeping bins written before CS-5 lacked `#[must_use]` — three latent layer-1 conservation gaps that the whole gate, 206 tests and five case-study reviews had never surfaced, because `must_use` was the one sealing rule no grep had gated (F-058). The linter found all three on day one; each was fixed with one attribute. A second measurement followed: a naive name-level audit of “does every tripwired type have a consumer?” reported 13 orphaned types, and resolving consumption through the bounds-first discipline (requirement satisfaction, supplier Taken lists, trait impls) found zero — any F-035 audit that does not follow the bounds reports false positives (F-059). The method’s instruments, once again, audited the method.

7.3 The DSL question: a mini evidence loop of its own

Step 10 — “a humane notation over the library” — was the programme’s longest-standing open wish, and it got the full treatment: three parallel experiments (EXP-13..15), one per candidate route, with error-message quality through the extra layer fixed in advance as the make-or-break criterion.

EXP-13, the macro front-end (`model!`), was rejected — but for the opposite of the expected reason. The feared failure, error quality, largely did not happen: `macro_rules!` only rearranges the modeller’s tokens, and spans follow tokens, so all three canonical violations kept their curated error head and tail identical to hand-written code (F-063). What killed the route is that a macro-authored model is *invisible to the R20/R21 toolchain*: run over the compliant DSL arm, docgen emitted 0 documents (control: 10), the context diagram collapsed to one node, and the linter raised two false gate-fatal errors while passing every sealing check **vacuously green** — it sees no resource types at all. An unauditable model passing the audit gate green is the F-037 silent direction at whole-model scale, in the gate built to prevent it (F-060). The savings (−44 % declaration-layer lines) never reached the interesting constructs, which lived in raw-Rust escape hatches anyway.

EXP-14, the external notation (`.lav` compiled to a committed crate), passed its make-or-break outright: generated code is ordinary source, so both error layers survive — conservation E0080s with the curated message, REQ-phrased E0277s — with a breadcrumb comment on each generated line making rustc’s “while instantiating” note render the notation line *inside the error*; and the real `trace.sh`, linter and docgen all run green over the output, because the emitter is a codification of the house style (F-061). The notation layer even adds errors Rust catches only at test time, reporting R2 double-use and R1 leaks at generation time with `file:line:col`. The honest cost: 113 notation lines replace ≈ 455 lines of Rust ($\sim 4\times$ denser) over *only* the continuous-resource core, and every further feature — discrete suppliers, fallibility, qualifications, money — grows grammar, checker and emitter together, regrowing the language it fronts (F-064).

EXP-15, generation from the specification itself, measured the split: the agreed `SPEC_TEMPLATE.md` format mechanically determines about half a model — 51 % by line count, 63 % of model items fully — and it is exactly the *structural* half: sealed state types, boundary impls, process signatures with conserving bodies and balance asserts, two passing flow tests. What it cannot determine, emitted as 12 compiler-enumerable holes, is exactly the half the method’s curated diagnostics live in: requirement bounds, characteristic consts, `Satisfies`: placement, all tests (F-062). The by-product proved the most immediately valuable result of the batch: *spec validation* is its own error class — balance arithmetic, waste-destination closure, name and id closure are all checkable against the document in milliseconds, where today an unbalanced spec line surfaces days later as an E0080 in the

implemented crate.

7.4 The verdict made rule: the R22 pipeline

R22 fixes the pipeline the trio points to. **Specifications are machine-validated before implementation:** `tools/spec.sh` (the `speccheck` binary of `tools/spec-gen`) runs as `ci.sh` step 8 over every case-study spec, with errors phrased at the document (“fix the specification, not the model”) and §/line references; the template gained the ten amendments (A1–A10) that make it mechanically parseable, and all five agreed specs were migrated. **New models can start as scaffolds:** the sibling `specgen` binary emits the spec-determined structure one-shot, every under-determined decision an enumerable `compile_error!` hole (`cargo build --features deny-holes` lists them all), and the output is promoted to hand-maintained in the same change that commits it — no regeneration round-trip, because EXP-15’s mechanical naming drift makes one fatal. **Model source is only ever ordinary Rust source:** a whole-model macro front is never built (F-060); macro sugar stays confined to model-core’s kernel macros — grown by exactly one declaration-layer family in this step, the boundary source/sink/entry generators, taught to the shared scanner in the same change and keeping the modeller’s tokens literal so spans and curated messages survive (F-063). The full external notation is deliberately **parked on the experimental track:** validated end to end (F-061), economically honest (F-064), and not adopted until a model comes along whose continuous-resource core would repay a regenerating crate.

8 Findings and results

Sixty-four findings condense into eight lessons.

8.1 What the compiler genuinely guarantees

At type-check time — visible in editors, in `cargo check` and to `trybuild` — the compiler enforces: exclusivity (a resource in use by one process at a time, F-025); no duplication and no creation from nothing (sealed types, F-005/F-006); state correctness (a product used before the process that makes it, a failure state continuing a success flow — F-023, F-047); unit and currency segregation (E0308, F-051); requirement satisfaction (REQ-bounded processes, F-044); capacity on discrete suppliers and consumers (an empty box does not implement `Supplier`, F-015); transposition in catalogues (kind traits, F-018); and wrong-price payment (exact-price vendors, F-051). These are the hard wins: they need no tests and no coverage, only the sealing checklist.

8.2 The post-monomorphization boundary

Everything arithmetic — mass balances, energy balances, overdraw of containers and budgets, per-branch conservation of fallible processes, exchange-rate exactness — is a genuine compile error that fires *late*: at monomorphization, invisible to `cargo check`, editors and `trybuild` (F-001, F-003, F-027, F-045). The project treats this as a first-class boundary with its own architecture: CI gates on full builds; conservation regressions are `rustdoc compile_fail` doc-tests; every process must be instantiated by a test; and the error-reading guide teaches that the real call site of a cross-crate E0080 is in the “while instantiating `fn draw_gas::<6000, 0, 5000>`” note, not the primary span. One genuine consolation, measured in EXP-09: unlike Peano capacities, `const-generic` magnitudes print as *decimal numbers* in every diagnostic, so continuous-resource errors are the most readable in the project (F-027). CS-4 added two refinements at multi-crate scale (F-001 extended): change impact is enumerated in per-crate

waves, because cargo build stops at the first failing crate; and a *non-generic* top-level `const _: () = assert!(...)` item evaluates at check time, so whole-model arithmetic hoisted into concrete `const` items is editor-visible after all — a zero-cost partial mitigation of the boundary.

8.3 The layered regime for silent loss

“Nothing is silently lost” is a property of the regime, not the type system (F-002): compile-time protection for discard-by-statement, `let _`, `forget` and explicit drop; test-time protection (the tripwire) for used-then-dropped values, forgotten match-arm fields and abandoned end states (the empty gas bottle, the zero-change `Money<0>` — F-032); and two accepted residuals — untested branches (branch coverage *is* leak coverage, R5) and panic unwinding, under which a bundle of five resources was demonstrated to vanish with no abort and no report (F-047). Three structural corollaries surfaced only in real builds: whoever mints a tripwired type must also ship a production-legal consumer for it, or downstream flows literally cannot account for it (F-035); contents-keeping consumers need a sealed recursive disposal path, or emptying the bin fires every kept tripwire (F-039); and tripwired wrappers must hold untripwired contents (F-040).

8.4 The SIGBUS, and errors as a designed surface

The project found one way to kill the compiler outright: a contents-keeping consumer with no decreasing space parameter diverges trait resolution, which at the default recursion limit is a graceful E0275 — but at the project-mandated `recursion_limit = "2048"` is a deterministic rustc SIGBUS with no diagnostic at all (F-034). The shape is now a hard rule (every contents-keeping consumer has a decreasing space parameter; R16’s type-level-recording History variant is forbidden for the same reason), and “compiler dies with no error” has a documented first suspect.

More broadly, error-message quality was treated as a first-class engineering concern — the errors *are* the product’s user interface — and the findings record both directions. Curated: `#[diagnostic::on_unimplemented]` turns capacity and requirement failures into sentences (“`BoltBox<Nil>` cannot supply anything: it is empty”; “this person may not drill: `Person<5000>` is not a certified drilling operator (REQ-004)”) — with the measured caveats that the attribute only fires on the trait-bound path (F-015), only the root obligation’s message is shown (so it must sit on the requirement trait, F-044), and the message is fixed per trait, not per impl. Uncurable and documented instead: aliases are erased (N41 vs N42 is two 40-deep `Succ` chains to count by hand), long types spill to side files from capacity ~130, duplicate bound errors appear at one call site (F-009), and composed processes emit echo E0080s whose notes point inside the library (F-051). The shipped error-reading guide, the deliberately Debug-less bundles and the “ignore the fix-its” rule are all consequences of taking this surface seriously.

8.5 Measured costs

The approach’s costs are quantified rather than waved at. **Compile time** is a non-issue at the intended scale: ~2.6 s for a full capacity-100 supplier cycle, ~1 s at Peano $N=100$, superlinear only past $N \approx 500$ (4.6 s at $N=1000$) — the guidance is capacities $\leq \sim 500$. CS-4 confirmed the flat region at real composite depth: the 25-cycle batch recursion over a 100-bolt box compiles in ~1.0 s, and the whole cold gate grew only 1.4% when its two crates joined (F-011 extended) — though the measured recursion-limit minima (≈ 151 for a 100-item supplier with its fill machinery) mean the default limit no longer covers what F-010’s bare $\approx N+3$ suggested, and test crates need the mandated attribute too. **Line counts**: the parameterized catalogue costs

33 lines where hand-written markers cost 121 at 24 combinations, and grows by 2–3 lines per new value; a sealed resource is one macro line (after F-036’s resolution); a hand-sealed wrapper like `Qualified` is ~34 lines once, versus ~40 duplicated lines *per type* on the rejected parallel-person design (F-043); `trace.sh` is ~150 lines of shell. **Const-parameter burden** is the real ergonomic tax: output magnitudes cannot be computed on stable, so callers state every total (F-022), budget parameters infect every signature their resource passes through with the running balance re-derived by hand (F-030), and a fallible process carries both branches’ splits — eight const parameters on the pilot’s drilling step, `drill_fallible::<1000, 4000, 5000, 450, 440, 10, 430, 20>` (F-045). None of this is a soundness cost — a wrong number anywhere is the E0080 — but it is typing, and it was the strongest argument for a DSL; the step-10 experiments measured what a notation actually buys against it (§7.3) and the answer settled the question at scaffolding, not notation.

8.6 Negative results that shaped the design

Several rejections are as load-bearing as the adoptions: binary type-level numbers (scale beautifully, cost 3× the machinery, and the capacity lists are $O(N)$ -deep anyway — F-012); aggregate resource bundles (they over-claim, deleting the concurrency R9 exists to allow — F-024); markers on `Person` (certifies everyone at once — F-043); type-selected outcomes (F-042); shadow lints (they fire on the conservation-correct idiom — F-007); and `Supplier` for continuous material (E0207 makes it impossible, and the packet workaround makes non-multiple amounts unreachable — F-028).

8.7 What multi-crate models and tooling added

The ladder’s last rungs produced a cluster of findings that only exist at subsystem scale. **Budgets and recursion** (F-056): a const time budget cannot descend through a type-level batch recursion on stable Rust; the working encoding is the quantum clock — structural Peano decrements of fixed-size quanta, with each draw minting conserved milliseconds — trading ad-hoc draw sizes for check-time, editor-visible overdraw errors. **Cross-crate money** (F-057): a downstream crate holds another crate’s sealed currency by *wrapping* it as payload (balance parameter and held cash equal by construction), and owning a currency now carries a stated obligation to ship parameterized boundary entry/exit functions, extending F-035’s “whoever mints ships a sink”. **Permits stop at the crate edge** (F-054 extended): permit-gated extraction is same-crate only, so cross-crate consumption is by keeping, with the sealed characteristic beside the type and the requirement trait beside the requirement’s owner. And **machine-reading the models** (F-055, extended across three builds) catalogued seven gaps between the grep discipline — which is sufficient for humans and `trace.sh` — and what *tools* need to recover full model structure from source: hidden supplier item types, where-clause bounds, many-to-many *satisfies!* resolution, path-qualified returns, hard-coded kernel semantics, flow structure hidden by statement-order freedom, and per-crate scanning that cannot see cross-crate flows. Most of the catalogue is now closed: the many-to-many ambiguity became a real silent diagram error in CS-2 and the generator now draws one edge per satisfying type (point 3 resolved), CS-3 adopted the match-directly-on-the-call convention for gap six, and the step-9 build resolved supplier-item recovery (gap 1) and taught the shared scanner to load workspace dependencies transitively (gap 7) — multi-crate diagrams now come out whole. What remains is conservative by design, surfacing as warnings rather than silent gaps.

8.8 What building the toolchain taught

Steps 9–11 added their own findings cluster (F-058..F-064; the prose account is §7). The pattern across all seven: **a gate is only as good as what its tools can see**. The one ungated sealing rule was the one silently broken in three places (F-058); a sink audit that does not resolve through the bounds discipline reports pure false positives (F-059); a macro-authored model passes every source-parsing check vacuously green (F-060) — which is why R22 fixes model source as ordinary Rust, with macro sugar confined to kernel macros whose token-preserving shape keeps the curated error surface (F-063). On the generation side: emitting ordinary source preserves both error layers and the whole tool surface (F-061); the agreed spec format mechanically determines the structural half of a model and none of its error-quality machinery, with the gap enumerable as holes rather than guesses (F-062); and a regenerate-forever notation must own everything it covers, so its economy ($\sim 4\times$ over the continuous core) shrinks as coverage grows (F-064). The adopted pipeline — machine-checked spec, one-shot scaffold, hand-written Rust — is these seven findings composed.

9 Limitations

The project’s credibility rests on stating these plainly; they are the documented record, not concessions extracted by critics.

- **Conservation errors are invisible while editing** (F-001, F-003). The balance checks fire only on a full build. `cargo check`, `rust-analyzer`’s default diagnostics and `trybuild` all pass violating models; generic code never instantiated is never checked. The mitigations (§4.4) are real but are mitigations: the editor is lying about balances until the build runs.
- **Silent loss is not fully closed** (F-002). The layered regime leaves two holes: a leak on a branch no test exercises is caught by nothing, and panic unwinding evades every layer silently. A panicking model run has already failed, but its accounting is void.
- **The boundary is discipline, not construction** (F-005). Sealing is one attribute away from broken — one `#[derive(Clone)]`, one `pub enum` — with no compiler warning; the defence is a greppable checklist and CI, and `unsafe` anywhere downstream voids it (hence `forbid(unsafe_code)` as policy).
- **Scale ceilings are real** (F-009, F-010, F-011). Type-level capacity costs `recursion_limit` $\approx N+3$ per crate and goes superlinear past $N \approx 500$; magnitudes are unusable as unary types; diagnostics erase aliases and spill long types to side files. A box of 100 bolts is fine; a warehouse is not a type.
- **One shape crashes the compiler** (F-034). Deterministically, with no diagnostic, at the mandated recursion limit. It is fenced by rule, not fixed.
- **The arithmetic burden is the modeller’s** (F-022, F-030, F-045). Every remainder and total is caller-stated; the compiler checks but never computes. Fallible processes roughly double the load.
- **Traceability is grep with a discipline** (F-021, F-037, F-038, F-053) — and two of its failure modes are *silently green* in `trace.sh` itself: a `Satisfies:` tag inside a macro invocation is dropped without warning, and a reused requirement id across crates merges reports while the gate stays green. Both now have dedicated gates layered on top — the linter’s macro-tag and duplicate-id ERROR checks (§7.2) and the specification gate’s workspace-unique id check (§7.4) — but the underlying grep remains structurally unaware of either shape.
- **Verification is not validation**. Tests prove the model self-consistent; nothing checks it against reality — CS-1’s energy figures are round-figure stand-ins, and every outcome-token constructor is a placeholder until calibrated (open question 1).

- **No tolerances, no time, no rates.** Quantities are exact integers — “1500 g $\pm$ 10 g” is inexpressible (open question 4); scheduling, duration and power are deliberately out of scope (R9, R15), so the History is a partial order, not a timeline.
- **The modeller must currently be Rust-literate.** The tutorials and the modeller’s guide lower the ramp, and the R22 pipeline now starts a model from a machine-checked specification and a generated scaffold — but the half of a model the spec cannot determine (§7.3) is hand-written Rust, and the full notation that would remove that is deliberately parked (§7.4): validated by experiment, not adopted.
- **Scope of evidence.** Everything above was measured on one toolchain (rustc 1.98.1, one platform) by one small team. The case-study ladder (§6) has now stressed the method from a pot of tea to a three-crate two-site model, but the largest model remains modest (ten crates, a 25-cycle batch); independent replication and industrial scale remain untested.

10 Related work

This project sits deliberately between several established fields; the debts are real even where the approach differs. No citations are given beyond names — the comparisons are general, and the repository makes no claim to a literature survey.

MBSE and SysML. Model-based systems engineering as practised — SysML and its successors, tools in the Capella tradition — gives systems a structured graphical language, with consistency checked by the modelling tool to the degree the tool implements it. Lavoisier differs in where the authority lives: the model *is* source code, and the checker is a production compiler maintained by a large community, with conservation, exclusivity and traceability as type-system theorems rather than tool-side validations. The cost is the loss of MBSE’s visual language and ecosystem — which steps 8–9 of the plan existed to win back from the code side, and now have: the generated three-level diagrams (docs/diagrams/, one set per model crate) and the generated process documents (docs/processes/, one per process — §7.1) are both committed, regenerated outputs of the models themselves.

Typestate and session types. Encoding a value’s legal state transitions in its type — *typestate* — is a long-standing idea, and Rust’s ownership model made “typestate pattern” a common idiom (builders that change type as they are configured). R9’s one-type-per-processing-state rule and R12’s capacity-indexed suppliers are typestate applied to physical things; the project’s contribution is scale and discipline rather than novelty, plus the measured record of where the diagnostics stop cooperating (F-009).

Linear and affine types. The “every resource used exactly once” ideal is linear typing, studied since Girard’s linear logic and implemented to varying degrees in research languages and in Haskell’s linear-types extension. Rust is affine (at most once), and F-002 is this paper’s quantification of exactly what that costs in practice — with the layered regime as an engineering answer where a linear language would have a theorem. A genuinely linear host language is the obvious alternative design; stable Rust was chosen for its maturity, diagnostics and reach.

Units-of-measure libraries. Checking dimensions in types is well-trodden: F# has units of measure in the language; Rust’s uom and C++’s mp-units do it in libraries. R7/R8 reimplement a small version deliberately (the experiment forbids dependencies, and the conservation asserts need the values, not just the dimensions); the per-currency dimensions of R19 are the same idea applied to money.

Formal methods. Specification languages with machine-checked consistency (the TLA+/Alloy/B tradition) prove far richer properties — temporal behaviour, refinement —

at the cost of a separate artifact and specialist skills. Lavoisier checks a narrower class of property (balance, exclusivity, state, capacity) but keeps the model, the implementation of the checks, and the documentation in one artifact a working engineer can run with two commands.

11 Future work

11.1 The planned deliverables: all landed

Every deliverable of the original plan now exists. Diagram generation (step 8), process document generation (step 9), model analysis (step 11) and the DSL verdict with its specification pipeline (step 10) are the subject of §7; each is a committed, regenerated output under its own gate. What the toolchain leaves open is deliberate: the external notation stays parked on the experimental track (EXP-14's `lavc` is evidence, not product) until a model arrives whose continuous-resource core would repay a regenerating crate — and its stated precondition, a notation-layer `--check pass pre-empting F-001`, is on record (F-061).

11.2 The case-study ladder

Complete: CS-1 through CS-5 are specified, agreed, implemented and gated, each stressing something no earlier one did — the full account is §6, and the per-case review decisions and round-trip feedback live in each `case-studies/*/SPEC.md`, all five now migrated to the machine-checked template and validated by the specification gate on every CI run. The next case studies will be the R22 pipeline's first real users: born as checked specs and `specgen` scaffolds, their enumerated holes filled by hand. What the ladder leaves deliberately untested is scale beyond the modest (a warehouse is still not a type) and models built by hands other than the project's own — the learning materials exist so that the next models need not be.

11.3 Open questions

Of the questions logged at the end of `instructions.md`: **validation vs verification** (calibration data for every placeholder constant and outcome token — the model is consistent; is it true?) remains fully open, and is now the largest one standing; **document generation** is answered and hardened into R20 (§7.1); **change impact and subsystem decomposition** has its systematic demonstration — CS-4's two probes and crate split (§6.4, `CHANGE-IMPACT.md`), with the caveats measured (per-crate waves; prose and the new numbers stay human work); and **tolerances** (exact integers cannot say ± 10 g; min/max const pairs could, at real complexity cost — currently a logged limitation, not a plan).

12 Conclusions

The hypothesis survives, with measured edges. A stable, dependency-free type system can carry a working systems-engineering method: resources that cannot be duplicated, shared or conjured; processes whose books must balance in every dimension they touch, including both arms of failure; requirements that are defined, satisfied and verified by machinery that fails the build when the chain breaks; and a boundary where every assumption is a named, greppable placeholder. Ten workspace crates — the library, the pilot and five case studies spanning a pot of tea to two-site fulfilment with procurement — pass an eight-step gate, and the first specification-to-model round trip found and fixed a flaw in the verification tooling itself (F-053). The completed ladder also converted the project's boldest convenience claim

into measurement: CS-4's change-impact exercise shows the compiler enumerating a probe's full blast radius — in per-crate waves, one error per source location however many cycles repeat it — while the human keeps the design decisions the errors demand (§6.4).

The edges are equally the result. The compiler's guarantees split cleanly in two: structural errors are caught everywhere, instantly; arithmetic errors are caught certainly but late, invisible to the editor. True linearity is out of reach — “nothing is silently lost” is a regime, not a theorem, and its two residual holes are documented. The ergonomic tax is real and quantified, from eight-const-parameter signatures to hand-maintained balances. And the project's most transferable lesson may be methodological rather than technical: agreeing requirements in writing, testing each risky claim in isolation before building, and logging every limitation with verbatim evidence produced a system whose weaknesses are as well-documented as its strengths — which is precisely what an engineering audience should demand of a method that proposes to audit *their* plans.

The paperwork question is now answered in both directions (§7): work instructions, diagrams and analysis reports are generated from the models, specifications are machine-checked before any code exists, and new models can begin as scaffolds whose every open decision is a named compile error. The notation that would finish the job is validated and deliberately parked. What remains — case studies born through that pipeline, and calibration against reality — is laid out in §11, and future versions of this document will report on each as it lands.

Appendix A: Glossary

Term	Meaning
Resource	A modelled thing (material, item, person, tool, container), represented as a sealed type that cannot be cloned, copied or constructed outside the boundary
Process	A function standing for a work step: takes everything it needs by value, returns everything it produces (R1)
Sealed type	A struct with at least one private field and no public constructor, derives or data-to-Self functions (F-005)
Boundary	The model's edge: the only place resources are created (suppliers, fill/draw functions) or finally leave (consumers, sinks) (R12)
Supplier / Consumer	Boundary traits for discrete entry/exit, with capacity in the type; strict (nothing but self and the item) (R12)
Draw process	The continuous-resource counterpart: takes a container, returns the drawn amount plus the container at its caller-stated remainder (R15)
Placeholder	A <code>/// Placeholder</code> : tagged boundary object or constant standing for an undecided real-world counterpart; the model's named assumptions (R12)
Tripwire	A Drop implementation on a consumable resource that panics at test time if the value dies unconsumed (R1 layer 2, F-008)
Defuse	The sealed, sanctioned <code>mem: :forget</code> site a consumer uses to accept a tripwired resource (F-008, F-032)
Conservation assert	An inline <code>const { assert!(...) }</code> balancing a process's inputs and outputs per dimension; violations are E0080 at monomorphization (R3, F-001)
Post-monomorphization error	A compile error that fires during code generation, invisible to cargo editors and trybuild (F-001, F-003)
Kind trait	A trait bounding a catalogue type's parameter slot (<code>Size</code> , <code>Material</code>), making transposed arguments a construction-site error (R6, F-018)
Requirement trait	A named trait <code>ReqNNN...</code> whose doc comment starts with its <code>REQ-NNN</code> id, used as a process bound; the R10 pattern
<code>Satisfies: / Verifies: trace.sh</code>	Greppable doc tags linking types/processes and tests to requirement ids; <code>Satisfies:</code> is backed by a compile-checked assertion (R10, F-020)
<code>ci.sh</code>	The dependency-free shell script producing the traceability report; any warning fails CI (R10, EXP-06)
<code>modellint / docgen</code>	The eight-step verification gate: full build, tests, clippy, plain production build, traceability, feature-placement audit, model analysis, specification gate (§4.4)
<code>speccheck / specgen</code>	<code>tools/diagram-gen</code> 's analysis and document-generation binaries, sharing one source-extraction library with the diagram generator (R20, R21)
<code>SPEC-HOLE</code>	<code>tools/spec-gen</code> 's binaries: the specification gate (<code>ci.sh</code> step 8) and the one-shot scaffolding generator (R22)
<code>test-support</code>	A numbered <code>compile_error!</code> a generated scaffold emits for every decision the specification under-determines; enumerated by cargo build <code>--features deny-holes</code> (R22, F-062)
	The cargo feature exposing downstream test fixtures; legal only under <code>[dev-dependencies]</code> (F-004)

Term	Meaning
Peano capacity	A type-level natural ($\text{Succ} \langle \text{Succ} \dots \langle \text{Zero} \rangle \rangle$) counting a supplier/consumer's remaining capacity (R12, EXP-01)
One type per state	Each processing, failure and safety state is its own type, converted only by processes (R9, F-023)
Outcome token	A sealed, runtime-valued token injecting a fallible process's outcome at the boundary; flows cannot read it (R17, F-042)
Outcome bundle	A $\#[\text{must_use}]$, deliberately Debug-less grouping returned per <code>Result</code> arm of a fallible process (R17, F-046/F-047)
History	The sealed boundary consumer recording consumed time and events: "the past" as a sink and the execution record, a partial order across branches (R16)
model-core	The reusable kernel library; the nine downstream model crates (the pilot and the eight case-study crates) depend on it
F-NNN / R-NN / EXP-NN / REQ-NNN	A finding; a rule of the modelling approach; an experiment; a requirement of a modelled system

Appendix B: The rules R1–R22, one line each

Rule	One-line statement
R1	Processes conserve resources: everything in by value, everything out, no side effects; resources have private constructors; the layered regime covers silent loss
R2	Strict conservation: resources are never borrowed or shared; reusables are moved in and returned
R3	Splitting and combining amounts are processes, balance-checked by <code>const</code> asserts
R4	Compile errors expose model mismatches; two error classes (type-check vs post-monomorphization); CI never gates on cargo check alone
R5	Unit tests verify every transformation — every branch of every process arm
R6	Characteristics are types: parameterized catalogues with kind traits; requirements always as trait bounds; one-way bridging
R7	Quantities and units are types carrying integer values in base units (g, mm, ms, mm ² , mm ³ , mK, mA, J, minor currency units)
R8	The project writes its own units library; no external units/number crates
R9	The model describes connections, not sequences; one type per processing state; loose threading by default
R10	Requirements traceability: REQ-NNN traits, tag + compile-checked assertion, tagged tests, trace.sh as a CI gate
R11	A shared library of common types (Person, Organisation, Location, units)
R12	Suppliers and consumers at the boundary: strict, finite, type-level capacity, Peano-counted, on_unimplemented-messaged; placeholders allowed and marked
R13	Discrete items are separate objects, never counts; fixed-size collections wherever possible

Rule	One-line statement
R14	Findings and limitations are logged in <code>FINDINGS.md</code> with stable F-ids and evidence citations
R15	Continuous resources are drawn-down containers; amounts, not rates; waste is an ordinary conserved output; unbounded sources/sinks only at the boundary
R16	Execution history: consumed time goes to the sealed <code>History</code> consumer; per-branch histories merged at joins into a partial order
R17	Fallible processes return <code>Result</code> with both arms conserving; boundary outcome tokens; <code>Debug-less</code> bundles; rework bounded by provisioning
R18	Qualifications and safety as types: the <code>Qualified</code> wrapper, one type per safety state, REQ-phrased refusals
R19	Money as a conserved dimension: one dimension per currency, exact-price vendors, boundary-only integer-exact exchange
R20	Process documentation is generated from the model, never hand-written: everything factual recovered from source, gaps stated rather than guessed
R21	Model analysis is one tool: per-crate lint reports with a severity ladder (<code>ERROR</code> / <code>ERROR-KNOWN</code> / <code>WARN</code> / <code>INFO</code>), gated with a staleness diff
R22	Specifications are machine-validated before implementation; model source is only ever ordinary Rust; generated source is deterministic, breadcrumb, hole-enumerating, and one-shot scaffolds promote immediately

Appendix C: Findings index by theme

Theme	Findings
Post-monomorphization visibility	F-001, F-003, F-027, F-045
Silent loss and the layered regime	F-002, F-007, F-008, F-032, F-035, F-039, F-040
Privacy and sealing	F-005, F-006, F-026, F-031
Type-level numbers, capacity, diagnostics at scale	F-009, F-010, F-011, F-012, F-013
Boundary machinery	F-014, F-015, F-016, F-028, F-029, F-034
Characteristics, catalogues, states, flows	F-017, F-018, F-019, F-023, F-024, F-025
Quantity ergonomics	F-022, F-030, F-033

Theme	Findings
Traceability discipline	F-020, F-021, F-037, F-038, F-053
Kernel macros	F-036 (resolved), F-040
Execution history	F-041
Fallible processes	F-042, F-045, F-046, F-047, F-050
People, qualifications, safety	F-043, F-044, F-048, F-049, F-054
Money	F-051, F-052, F-057
Machine-reading the models (tooling)	F-055 (points 1, 3, 7 resolved), F-059
Multi-crate models and scale	F-054 (ext.), F-056, F-057; F-001/F-010/F-011 extensions
The toolchain and the generation regime	F-058, F-060, F-061, F-062, F-063, F-064